120 Days of Quantitative Finance

Week 15: Advanced Time Series Modelling

Day 104: Forecasting Returns and Volatility

1 Introduction

Forecasting financial markets involves predicting both expected returns and future volatility.

These two components are essential for trading, risk management, and portfolio optimization.

2 Return Forecasting

Returns are often modeled using ARIMA-type models:

$$r_t = c + \sum_{i=1}^{p} \phi_i r_{t-i} + \epsilon_t$$

Key idea:

- Capture autocorrelation in returns
- Extract short-term predictive signals

3 Volatility Forecasting

Volatility is modeled using GARCH models:

$$\sigma_t^2 = \alpha_0 + \alpha_1 \epsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$$

Key idea:

- Volatility evolves over time
- Depends on past shocks and past volatility

4 Combined Framework

A common approach:

- ARIMA for mean (returns)
- GARCH for variance (volatility)

$$r_t = \mu_t + \epsilon_t, \quad \epsilon_t \sim N(0, \sigma_t^2)$$

5 Forecasting Procedure

1. Fit ARIMA model to returns
2. Extract residuals
3. Fit GARCH model to residuals
4. Forecast mean and variance

6 Financial Interpretation

- Mean forecast → expected return (signal)
- Volatility forecast → risk estimate

7 Applications

- portfolio allocation
- risk management (VaR)
- algorithmic trading

8 Conclusion

Forecasting returns and volatility provides a comprehensive view of market dynamics and is central to quantitative finance.